

PRANAVA HOMEFLOW 2.0

World-Class Post-Sales Customer & Unit Operating System

Full Product, Process, Experience, Data, Intelligence and Implementation Design Specification - v8 - Project-First, Unit-Aware, Configurable Journey and System-Independent Architecture

North-star promise

Every customer knows what is happening. Every employee knows what they own. Every manager knows where risk lies. Every commitment has accountability. Every unit has a permanent digital history. Every rupee and margin leak is visible.

Design target: premium customer experience + disciplined execution + controlled transparency + profitable growth

1. Executive Summary

HomeFlow should evolve from a strong post-sales workflow application into Pranava's Customer & Unit Lifecycle Operating System. The existing modules are a valuable base and should be retained. The next stage is to connect them through a common data model, universal action engine, digital twins, predictive risk, customer-facing transparency, evidence-based quality, and profitability controls.

The operating hierarchy is Project-first and Unit-aware. Every material booking, unit, collection, change request, handover, task, team assignment, KPI and forecast must resolve to a Project. Users may work on one Project or on a shared cross-project team, but reporting, permissions, workload and cash-flow views must always support Project as a first-class dimension.

The design has two permanent anchors: the Customer Digital Twin and the Unit Digital Twin. A Booking connects them. The Unit Digital Twin is live even before a unit is sold: it carries construction progress, component status, configuration, changeability gates and forecast gate-expiry dates. This lets Project/Design teams publish the physical truth once while Sales and CRM use the same truth to guide unit selection, customer customisation and post-booking execution. The operating system then orchestrates all events, money, documents, quality, commitments and communications around that relationship.

New v8 operating principle: HomeFlow uses a generic Pranava lifecycle template that each Project inherits and configures. Stage durations, planned dates, revised forecasts, actual dates, SLAs, dependencies, gates, owners and customer visibility are configurable data - never hardcoded in software. Project-specific SOPs such as East Crest should be mapped into this common engine, and the common engine should in turn improve and standardize those SOPs.

System independence principle

HomeFlow must be fully operable as a standalone post-sales operating system. No external construction, ERP, CRM, finance, document, communication or facility-management platform is assumed to exist. Project/Construction teams must be able to maintain the authoritative physical status of Projects and Units directly in HomeFlow. External systems, if adopted later, are optional accelerators connected through generic adapters/APIs/events; they must never be required for core HomeFlow workflows or dictate the HomeFlow domain model.



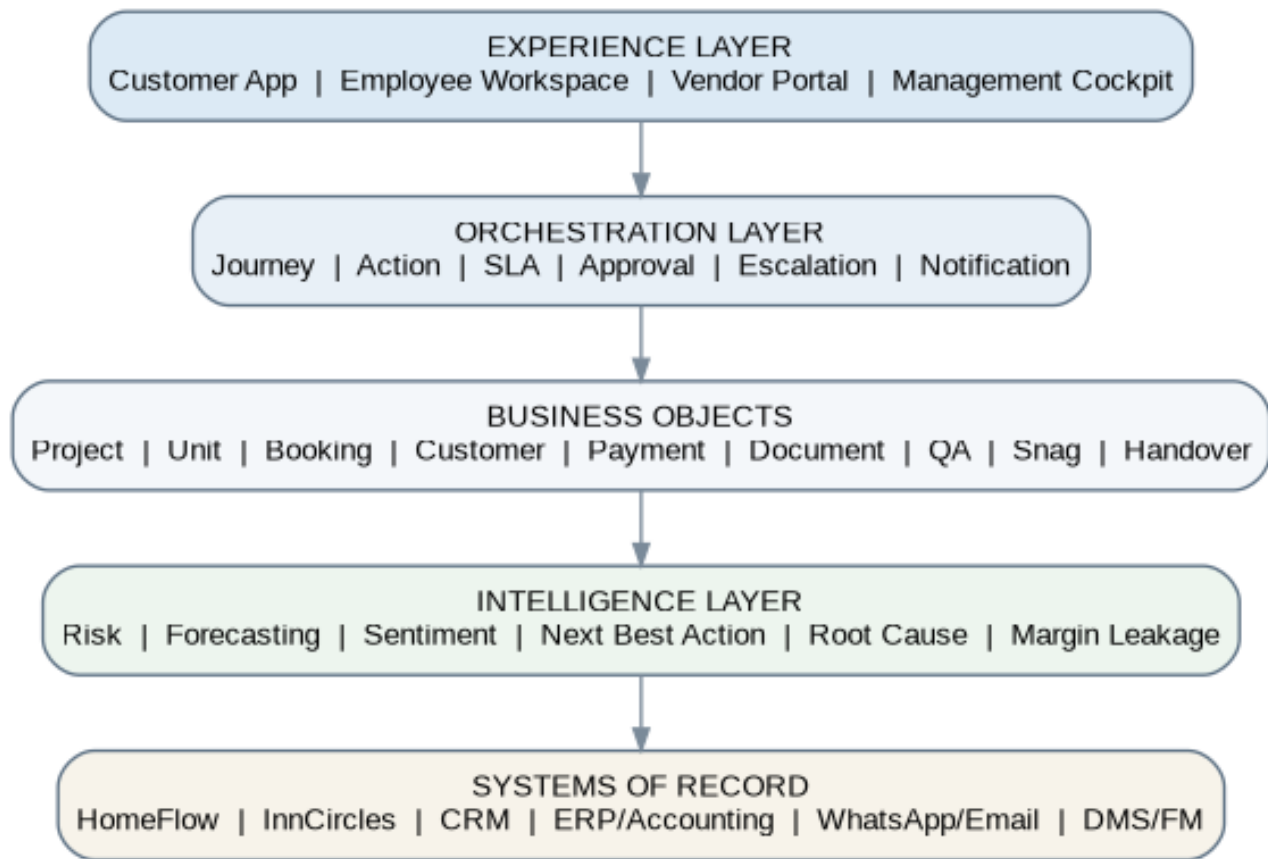
Outcome	What HomeFlow 2.0 must achieve
Experience	A premium, proactive and predictable journey from booking through post-handover.
Efficiency	Employees work from prioritized actions instead of searching modules and chasing colleagues.
Openness	Customers and stakeholders see the right information, at the right level, with clear ownership and timelines.
Control	Every promise, approval, SLA, blocker and exception is traceable.
Profitability	Collection risk, concessions, rework, delay cost and cost-to-serve are measurable by customer and unit.

Outcome	What HomeFlow 2.0 must achieve
Learning	Snags, delays, complaints and leakage feed root-cause analysis and process improvement.

2. Product Vision and Design Principles

- One truth, many views: the same customer/unit state powers customer, employee, vendor and management experiences.
- Customer + Unit, not module-first: screens and workflows organize around lifecycle outcomes, not departmental silos.
- Project -> Phase/Tower/Block -> Unit is the primary physical hierarchy. Customer/Booking and operational records attach to this hierarchy so every analysis can roll up from Unit to Project and across the portfolio.
- Manage by exception: management sees risks, breached commitments and forecast slippage, not raw task lists.
- Predict before failure: detect conditions that create delay, default, escalation, rework or margin leakage.
- Human judgment where it matters: automate routine steps, preserve human approval for consequential decisions and sensitive customer communication.
- Evidence over declarations: readiness, QA and completion require checklists, evidence and verification, not manual percentages.
- Controlled openness: transparency is role-based, permissioned and purpose-specific.
- Every action has an owner, SLA, dependency, business impact and evidence trail.
- Every important customer-facing promise is captured and monitored before it breaks.
- Every defect, delay and concession becomes data for future prevention and profitability improvement.
- Changeability is a live Unit property: Project/Design own the physical truth; Sales/CRM consume it but cannot alter it.
- Parallel streams, not a single waterfall: construction, finance, documentation, customisation, commitments and communication can run concurrently while milestone gates coordinate convergence.
- SLA is not the same as plan: SLA defines allowed service time; planned dates define when a specific record is expected to occur. Both must be measurable independently.
- Plan -> Current Plan -> Forecast -> Actual: every meaningful stage, sub-stage, milestone and task supports baseline planning, revisions, forecast dates, actual dates and variance analysis.
- Generic core, project-specific configuration: Pranava defines a standard lifecycle template; Projects inherit it and may apply controlled, effective-dated overrides without code changes.

3. Target Architecture



HomeFlow should act as the orchestration brain. It should not duplicate deep construction management, accounting, banking or document-storage systems when those systems already exist. It should consume authoritative data, convert it into lifecycle state, trigger actions, expose controlled transparency, and maintain the customer/unit audit trail.

4. Core Data Model: Customer + Unit + Booking

Entity	Purpose	Must persist independently?
Project / Phase / Tower	Physical and commercial hierarchy; policies, milestones and common metadata.	Yes
Unit	Permanent property identity and physical lifecycle.	Yes
Booking	Commercial relationship connecting one unit to one or more customers for a defined ownership period.	Yes
Customer	Person/entity relationship record, preferences, communication and experience history.	Yes

Entity	Purpose	Must persist independently?
Applicant / Co-owner	Party participating in a booking.	Yes
Lifecycle Event	Dated event that changes state or records evidence.	Yes
Action	Universal work item generated manually, by workflow, rule or AI.	Yes

4.1 Unit Digital Twin

- Identity: project, phase, tower/block, floor, unit number, type, area, facing, parking, UDS/land share where applicable.
- Specification baseline: standard finish schedule, brands, fixtures, approved drawings, customer change requests, approved variations, superseded versions and final as-built deviations.
- Construction state: milestone progress imported from project/construction systems.
- Component-level progress: room/trade/system state sufficient to determine whether a requested change is still practical.
- Live changeability state by change category/component: Open, Closing, Conditional, Exception Only, Hard Closed.
- Gate-expiry forecast: expected closure date/time and event that will close or restrict each change category.
- Changeability score / customisation flexibility index for sales inventory discovery, always explainable by underlying gates.
- Current released configuration: baseline specification plus approved variations and latest controlled drawing/specification revision.
- QA/QC evidence: component checklists, inspections, photographs, tests, certificates, approvals.
- Snag history: defect, severity, trade, contractor, root cause, cost, before/after evidence, repeat flag.
- Handover evidence: readiness gates, meter readings, keys, manuals, final photographs and signatures.
- Digital Home Passport: equipment, serial numbers, warranty documents, manuals, paint/tile codes, service history.
- Permanent history across ownership changes, cancellations, transfers and future resale.

4.2 Customer Digital Twin

- Profile and relationship: applicants, contact preferences, language, NRI/resident flags, relationship history.
- Bookings and unit relationships, including historical/cancelled/transfer records.
- Financial behaviour: demands, receipts, overdue history, payment promises, disputes, TDS and loan dependence.
- Documents and compliance: KYC, agreements, loan documents, registration documents, missing/discrepant information.
- Commitments: every promise made by Pranava and every promise-to-pay made by the customer.
- Communications: call notes, WhatsApp/email/SMS, meetings, notices, customer-facing and internal communication.
- Experience signals: CSAT/NPS, sentiment trend, complaints, escalations, referral/advocacy status.
- Consent and privacy preferences.

4.3 Booking as the Bridge

4.4 Project as the Operating Partition

- Every Unit belongs to exactly one Project and, where applicable, a Phase, Tower/Block, Floor or Villa Cluster. These hierarchy keys must be mandatory on all Unit master records.
- Every Booking, Demand, Receipt, Loan Case, Change Request, Snag, Handover, Commitment, Escalation and actionable work item inherits Project from its linked Unit/Booking. Manual project selection should be avoided when it can be derived.
- Project is a universal filter and security dimension across dashboards, reports, work queues, forecasts and management cockpits. Portfolio views aggregate projects; users can always drill from Portfolio -> Project -> Unit -> Customer/Booking.
- Teams may be dedicated to one Project, shared across selected Projects, or centralized across the portfolio. HomeFlow must model team-to-project assignments with effective dates, roles, workload ownership and escalation routes.
- Project configuration may override portfolio defaults for workflows, SLAs, customisation policies, approval matrices, communication templates, handover gates and reporting targets without breaking the common enterprise data model.

Critical modeling rule

Do not attach all commercial and lifecycle data directly to the customer or directly to the unit. Attach it to the Booking wherever the fact belongs to a particular customer-unit ownership relationship. This prevents data corruption during cancellation, transfer, joint ownership changes and resale.

5. End-to-End Lifecycle Blueprint



Generic stage	Primary outcome	Typical parallel streams / exit logic
0. Unit / Pre-Sales Readiness	Sales has reliable unit truth before booking.	Project-owned physical progress, expected readiness and changeability visible to Sales/CRM. No customer required.
1. Booking & Allotment	Commercial booking accepted and customer relationship initiated.	Booking amount, application, allotment, applicants, commercial approvals, sales-to-CRM handover.
2. Funding & Financial Setup	Funding route established.	Loan sanction or self-funding plan, KYC/funding evidence, payment schedule; may continue in parallel with documentation.
3. Agreement & Documentation	Contractual package complete.	KYC, AOS/agreement generation, legal/commercial approvals, TDS/document obligations, execution

Generic stage	Primary outcome	Typical parallel streams / exit logic
		and archival.
4. Construction & Unit Journey	Unit progresses transparently against Project-owned physical truth.	Unit Digital Twin, component progress, evidence, customer updates, approved changes/customisations and future changeability gates.
5. Demands & Collections	Milestone-linked receivables are generated, collected and forecast.	Demand generation, loan disbursement, TDS, receipts, overdue recovery, promise-to-pay and project cash-flow forecast.
6. Pre-Registration Readiness	All prerequisites for registration converge.	Financial clearance, legal/document verification, statutory items, customer availability, approvals and registration gate.
7. Registration	Legal transfer/registration completed and evidence archived.	Sale deed/document generation, challans, slot scheduling, execution, registered copy and status updates.
8. Pre-Handover Readiness	Physical, commercial and customer readiness converges.	Unit readiness, QA, snags, financial/legal closure, commitments, customer inspection scheduling and hard/soft handover gates.
9. Handover & Possession	Property and required information are formally handed over.	Keys, possession documents, readings, manuals/warranties, customer acknowledgement, FM orientation and final evidence.
10. Post-Handover / Facilities	Customer transitions into living/occupancy and support.	Move-in, DLP/warranty, service requests, FM onboarding, home passport, satisfaction and advocacy.

6. Readiness and Health Scores

HomeFlow should avoid one ambiguous percentage. Scores must answer distinct operational questions and reveal their components.

Score	Question answered	Illustrative dimensions
Unit Readiness	Is the physical flat/villa genuinely ready?	Construction completion, QA checks, utilities, common-area dependencies, snags.
Customer / Booking Readiness	Is the customer-side relationship ready?	Payments, loan, KYC, legal, registration,

Score	Question answered	Illustrative dimensions
		commitments, scheduling.
Handover Readiness	Can we safely and successfully hand over now?	Weighted combination of physical + customer readiness + mandatory gates.
Customer Health	How healthy is the relationship?	Sentiment, SLA, commitments, complaints, payment behaviour, communication pattern.
Financial Health	How likely are collections to occur on plan?	Overdue, promises-to-pay, loan gaps, disputes, ageing, behavioural risk.

Every score must show: current value, trend, top three drivers, confidence level, and recommended actions. Scores must not become decorative badges.

7. Universal Action Engine: The Operating Backbone

Every actionable item in HomeFlow should normalize into one Action object, whether it originated from a task, snag, commitment, document gap, payment delay, approval, escalation or AI recommendation.

Field	Requirement
Action ID / type	Unique and typed: Task, Approval, Document, Payment, Snag, Commitment, Customer Contact, etc.
Related objects	Customer, Booking, Unit, Project and originating record.
Owner / backup owner	Named person and department, plus re-assignment history.
Priority	Operational priority plus customer/financial/reputation severity.
SLA / due date	Policy-driven and pausable only with approved reason.
Dependencies	Explicit upstream blockers and downstream impact.
Business impact	Customer impact, rupee exposure, milestone impact, reputation risk.
Evidence	Attachment, photo, approval, transaction or structured proof required for closure.
State	New, In progress, Waiting internal, Waiting customer, Blocked, Ready for approval, Closed, Cancelled.
Audit trail	Every change, comment, reassignment, breach and approval timestamped.

7.1 My Day: Employee Work Cockpit

Behavioral design goal

Employees should not search HomeFlow for work. HomeFlow should prioritize the work that matters today.

- Top actions ranked by deadline, customer impact, revenue impact, dependency impact and escalation risk.
- Why now? explanation in plain language.
- One-click execution: call, WhatsApp, request document, approve, assign, escalate, create recovery plan.
- Focus modes by customer, project or functional queue.
- Daily closure summary and carry-forward reasons.

8. Module-Level Functional Specification

8.1 Sales Handover

Make CRM acceptance a controlled quality gate.

- Completeness score before Sales can submit.
- Mandatory vs conditional document checklist by project/product/customer type.
- Commercial approvals and deviations attached to booking.
- First-time-right metric by salesperson/team.
- Return reason taxonomy and repeat-error analytics.
- Automatic CRM task generation on acceptance.

8.2 Documents & Document Generation

Turn document handling from repository work into an intelligent pipeline.

- Auto-classification and metadata extraction from uploaded documents.
- Name/PAN/address/unit/value cross-checks against master data.
- Discrepancy flags with confidence score and review queue.
- Data-driven document generation with minimal re-entry.
- Version comparison highlighting substantive changes.
- Generate -> validate -> legal approve -> customer approve -> eSign -> archive.
- Expiry/renewal alerts for applicable documents.
- Dedicated Legal Document Factory for Agreement of Sale (AOS), Sale Deed, Lease Agreement, Addenda, Possession/Handover documents, declarations, NOCs, letters and other approved project/customer documents.
- Template-driven generation from Project, Unit, Booking, Customer, commercial, registration and approved customisation data; eliminate duplicate manual entry wherever a trusted source field exists.
- Clause Library with conditional clause logic by project, property type, customer type, transaction type, payment plan, jurisdiction and approved deviation.
- Pre-generation validation: mandatory-data completeness, cross-field consistency, duplicate/conflicting values, amount/date checks, unit identity, applicant names, PAN/address and commercial approval state.
- Draft -> internal review -> legal approval -> customer review -> execution/eSign/registration -> final signed copy -> permanent archive, with immutable version history.
- Controlled redline/compare between template revisions and customer/legal negotiated versions; highlight only substantive changes for reviewer attention.
- Every generated document must remain linked to Project -> Unit -> Booking -> Customer and be visible from both the Unit Digital Twin and Customer Digital Twin.

8.3 Collections

Move from reporting outstanding to predicting cash realization.

- Separate outstanding, due, overdue, disputed, loan-dependent, promise-to-pay and true risk.
- Reason codes for every overdue amount.
- Collection forecast by 7/30/60/90 days.
- Promise-to-pay tracking with breach prediction.
- Interest/penalty waiver approval and leakage tracking.
- Reconciliation exceptions and TDS verification workflow.
- Portfolio heatmap by project, RM, ageing and risk.
- Project-wise cash-flow forecast for configurable periods: current month, next month, weekly, monthly, quarterly and custom date range.
- Actual vs Forecast vs Revised Forecast comparison by Project and portfolio, including last month / prior period actual collections and variance.
- Forecast waterfall by source: scheduled contractual demands, overdue recoveries, promise-to-pay, expected loan disbursements, registration/final-demand events and approved reschedules.
- Probability-weighted expected collection date and amount at Booking/Unit level; roll up to Project, RM/team and portfolio without losing drill-down.
- Separate committed receivables from optional scenario inputs such as future sales collections. Post-sales cash forecast must never silently mix booked receivables with speculative new-sales inflows.
- Cash-flow plan view should show opening receivable, demands becoming due, expected receipts, overdue recovery, loan inflows, likely shortfall, closing receivable and collection confidence.
- Forecast accuracy KPI: compare original forecast, revised forecast and actual realization by Project, team, collection reason and horizon; learn systematic optimism/pessimism.
- Scenario capability: Base / Conservative / Stretch forecast with transparent assumptions and no overwriting of the baseline forecast.

8.4 Loans

Prevent future overdue by managing disbursement readiness proactively.

- Sanctioned, disbursed, available balance and next demand.
- Expected bank release date and missing documentation.
- Days-to-demand vs days-to-disbursement gap.
- Lender contact and case history.
- Risk score and automated actions before demand becomes overdue.

8.5 Legal

Make legal work exception-driven and traceable.

- Agreement template and clause library by project/entity/customer type.
- Deviation register with approval authority matrix.
- Version comparison and material-change summary.
- Legal query / response SLA and blocker ownership.
- Approval evidence and immutable audit trail.

8.6 Registration

Make registration readiness explicit and forecastable.

- Readiness checklist covering documents, payments, TDS, appointments and signatures.
- SRO slot scheduling and change history.
- Pre-registration gate with hard blockers.
- Day-of-registration checklist and live exception handling.

- Registered document receipt, validation and archive.
- Forecasted registration date with confidence and critical path.

8.7 Customer Change Requests & Unit Customisations

Treat customer-requested changes as controlled variations to the Unit Digital Twin, not as informal commitments, comments or snag items. The workflow must protect customer choice while preserving technical integrity, construction flow, handover predictability and project margin.

- Single Change Request ID linked to Customer, Booking, Unit, Project and affected room/trade/system.
- Support multiple line items within one request, such as wall relocation, flooring upgrade, electrical additions, sanitaryware change, kitchen modification or landscape variation.
- Standard catalogue items vs bespoke changes clearly distinguished; standard options should use pre-approved scope, price and lead time wherever possible.
- Feasibility workflow: Requested -> Under Feasibility -> Feasible / Feasible with Conditions / Rejected.
- Mandatory impact assessment before customer approval: Cost, Schedule, Technical/Design and Handover impact.
- Change-freeze date by project, tower/villa, trade and construction milestone, with post-freeze exception approval.
- Customer quotation/version with scope, inclusions, exclusions, drawings/specification, tax, payment terms, lead time and validity date.
- Execution release only after required approvals and configured payment gate are satisfied.
- Revised drawing/specification version control with superseded-document lockout and controlled release to Site, QA, Procurement and Contractor/Vendor.
- Automatic generation of site/procurement/vendor actions and dependencies from the approved variation.
- QA verification and before/after evidence before customer acceptance and closure.
- Completed change updates the Unit Digital Twin and as-built Home Passport permanently.
- Cancellation/reversal workflow with cutoff rules, abortive-work cost, refund/credit treatment and approval trail.
- Full profitability view: customer price, vendor/contractor cost, internal incremental cost, tax, discount/waiver, contribution and margin leakage.

Recommended end-to-end status flow

Draft / Requested -> Feasibility Review -> Costing -> Awaiting Internal Approval -> Awaiting Customer Approval -> Awaiting Payment -> Approved for Execution -> Released to Site -> In Progress -> Ready for QA -> QA Verified -> Customer Accepted -> As-Built Closed. Rejected, Withdrawn and Cancelled states remain auditable.

Core control model

Control dimension	HomeFlow requirement	Why it matters
Request definition	One change request may contain multiple line items by room/trade, with customer intent, drawings/photos, priority and desired date.	Prevents informal messages from becoming uncontrolled site instructions.
Feasibility	Design/technical review determines feasible, feasible with conditions, or rejected, with reason and	Protects structural, MEP, statutory, waterproofing, fire/life-safety and maintainability requirements.

Control dimension	HomeFlow requirement	Why it matters
	dependencies.	
Commercial control	Estimate customer price, incremental internal/vendor cost, taxes, discount/waiver, payment terms and gross contribution before approval.	Protects margin and avoids execution before commercial closure.
Schedule impact	Record lead time, procurement impact, construction impact, critical-path impact and handover-date impact.	Stops a small-looking change from silently creating a large delay.
Approval matrix	Authority depends on change type, technical risk, value, margin, schedule impact and whether the request is before/after freeze date.	Creates controlled flexibility instead of ad-hoc exceptions.
Customer acceptance	Customer receives scope, drawings/specification, price, tax, exclusions, schedule impact and validity date; explicit acceptance is captured.	Creates transparent informed consent and reduces future disputes.
Payment gate	Configured advance/full payment must be received before release for execution, unless an authorized exception is recorded.	Prevents unfunded variations and receivable leakage.
Drawing/spec control	Approved revision receives a version number; superseded drawings are locked and visibly obsolete; Site/QA/Procurement consume only released revision.	Prevents execution from the wrong drawing or specification.
Execution + QA	Released change generates site/procurement/vendor actions, evidence requirements, inspection and QA verification.	Connects customer request to physical delivery and quality.
As-built closure	Completed variation updates Unit Digital Twin, final drawings/specifications, asset/warranty records and handover pack.	Preserves the permanent truth of what was actually delivered.

Experience rule: the customer should be able to see exactly what was requested, what Pranava has approved, what it costs, whether it affects handover, the latest approved drawing/specification, what is being executed and what has been delivered - without exposing internal vendor pricing or internal deliberations.

8.7.1 Unit Progress, Configuration & Changeability Engine

HomeFlow must treat construction and sales as parallel processes. Units may be sold before construction starts, during active construction, near completion, or after physical completion. Therefore customisation availability cannot be derived from booking date or one project-wide cutoff. It must be derived from the live physical state of each Unit Digital Twin and governed by configurable gates.

Operating principle

A Change Request may be raised at almost any point in the lifecycle. Gates govern feasibility, approval, execution, commercial treatment and required authority; they should not unnecessarily suppress request creation. Sales and CRM may read the current changeability state. Project/Construction, Design and authorized rule owners control the physical state and gate logic. Safety/statutory hard gates are never overridden.

Changeability state	Meaning	Sales/CRM treatment
OPEN	Change can normally be evaluated and executed without rework if technically feasible.	Show as available; customer may shortlist based on this capability.
CLOSING	Still open, but a known construction/procurement event is expected to close it soon.	Show expected closure date/event and remaining window.
CONDITIONAL	Possible, but creates additional technical, procurement, cost or schedule conditions.	Show conditions; no promise until feasibility/quotation is approved.
EXCEPTION ONLY	Normal window has closed; execution may require rework or senior approval.	May be requested; clearly flag as exception and likely cost/time impact.
HARD CLOSED	Not permissible because of structural, statutory, fire/life-safety, sanctioned-plan or other non-overridable rule.	Do not offer as available; explain reason category without exposing internal noise.

Project / Construction progress interface

- Provide one Unit Progress Control interface with Project -> Phase/Tower/Block -> Floor/Zone -> Unit drill-down.
- Project team records actual physical states (for example: masonry not started / in progress / complete; MEP first-fix not started / active / complete; flooring not ordered / ordered / delivered / installed). The system derives change gates from rules wherever possible.
- Support bulk progress updates by tower/floor/zone/work package with preview of affected units and gate transitions before commit.
- Allow unit-level exceptions where actual progress differs from the bulk construction state.
- Show source and timestamp for every state: HomeFlow Project-team update, approved bulk update, QA evidence, procurement/material update recorded in HomeFlow, authorized correction, or an optional generic external integration if one is configured.
- Changes to a physical state that reopen a previously closed gate require a reason and audit trail.

Sales / CRM live inventory experience

- Sales inventory must expose Construction %, expected possession/handover window, Customisation Flexibility score and the current gate summary for every available unit.
- Filters should include Highly Customisable, Layout Flexible, Kitchen Changes Open, Electrical Changes Open, Flooring Selection Open, Bathroom Specification Open and Ready-to-Move.
- Sales can compare units side by side against a prospect's stated personalisation requirements before booking.
- CRM sees the same changeability state when managing a booked customer and cannot promise a change that is not currently Open without creating the required feasibility/exception workflow.
- Every displayed gate must include source, last-updated time and confidence/freshness indicator so Sales can distinguish current information from stale project data.

Requirement-to-unit matching

During sales discovery, capture the prospect's personalisation needs as Must Have / Preferred / Not Important. HomeFlow should match these requirements against available inventory and live unit gates, producing a Requirement Compatibility score with a plain-language explanation. The recommendation must never imply engineering approval; it only indicates current changeability potential.

Example	Unit V101	Unit V104
Construction state	Early construction	Near completion
Customisation flexibility	92/100 - High	12/100 - Minimal
Kitchen layout	Open	Exception Only
Electrical additions	Open	Conditional
Flooring selection	Open	Hard/Commercially Closed per policy
Sales interpretation	Strong fit for a customer seeking personalisation.	Strong fit for a customer prioritising early possession, not major changes.

Gate-expiry forecast and controlled hold

- For every Closing gate, show the expected closure date/time, closing event, and source schedule.
- Allow a tightly controlled Change Window Hold for serious sales opportunities when a construction activity is about to close a requested option.
- A hold must specify Unit, gate/category, prospect/opportunity, requested duration, construction impact and approver. It must automatically expire and release.
- Project/Construction approval is mandatory for any hold that changes planned execution. Sales cannot create a binding hold unilaterally.
- Configure maximum hold duration, number of concurrent holds, value/role thresholds and blackout activities where holds are prohibited.

Gate derivation examples

Physical / procurement event	Illustrative automated gate transition
MEP first-fix not started	Electrical / plumbing point relocation = OPEN

Physical / procurement event	Illustrative automated gate transition
MEP first-fix commenced	Relevant MEP changes = CLOSING or CONDITIONAL
MEP first-fix completed / wall closed	Relevant MEP relocation = EXCEPTION ONLY unless policy makes it HARD CLOSED
Flooring selection not frozen and PO not released	Flooring selection = OPEN
Flooring PO released	Flooring selection = CONDITIONAL
Flooring installed	Flooring replacement = EXCEPTION ONLY
Structural element cast / approved structural stage passed	Structural alteration = HARD CLOSED
Customer change approved and released	Affected configuration becomes controlled current revision; competing change rules re-evaluated

Permissions and source-of-truth ownership

Role	Authority
Project / Construction	Own actual physical progress and planned activity dates; approve schedule-impacting holds.
Design / Engineering	Own technical change rules, hard constraints and feasibility decisions.
Procurement	Provide PO/order/material status that can restrict changeability.
Sales	Read unit changeability, capture prospect requirements, compare units, request controlled holds; no gate editing.
CRM / CX	Raise/manage booked-customer change requests and customer communication; no physical-state editing.
Commercial / Finance	Own variation pricing, discount/waiver authority, payment gate and contribution view.
QA	Own inspection/verification states used for readiness and closure.
Management / Named Authority	Approve configured exception overrides; cannot override hard safety/statutory gates.

8.8 Unit Readiness & QA/QC

Replace subjective percentages with evidence-based completion.

- Component hierarchy by room/trade/system.
- Checklist-based completion with mandatory photographs/tests/certificates.
- Site declaration and independent QA verification as separate states.
- Common-area / utility / statutory dependencies linked to unit eligibility.
- Readiness score derived from evidence, not manual entry.
- Exception queue for failed inspections and repeat failures.

8.9 Snagging

Close the loop from defect discovery to verified closure and root cause.

- Snag category, severity, location, trade, vendor/contractor and root cause.
- SLA by severity and customer impact.
- Before/after evidence and QA verification.
- Customer verification where appropriate.
- Repeat defect flag and cost of rectification.
- Analytics by contractor, trade, project, root cause and cost.

8.10 Handover

Operate handover as a gated readiness event, not an appointment alone.

- Unified Handover Readiness Score.
- Mandatory hard gates: financial, legal, registration, QA, critical snags, commitments, FM/utilities.
- Predicted handover date and confidence.
- Appointment workflow and customer confirmation.
- Digital handover checklist: keys, meters, manuals, warranties, signatures, photographs.
- No override of hard gates without named authority and recorded reason.

8.11 Customer Commitments

Create a permanent Promise Ledger.

- Promise, owner, beneficiary, due date, financial impact, approval and evidence.
- Internal commitment vs customer-facing commitment distinction.
- AI-assisted promise detection from logged communication.
- Confidence score based on dependencies.
- Pre-breach alerts and recovery plan.
- Broken-promise rate by team and root cause.

8.12 Communications

Create one auditable communication history without exposing internal chatter.

- Omnichannel log: call, email, WhatsApp, SMS, meeting and notices.
- Strict separation of internal and customer-visible content.
- Templates with project/legal/compliance approval.
- AI conversation summary and open-decision extraction.
- Sentiment trend and unanswered-question detection.
- Communication frequency guardrails to prevent spam.

8.13 Escalations

Turn escalation into an exception-management engine.

- Rule-based escalation by SLA breach, severity, financial impact, commitment risk and sentiment.
- Tiered escalation path with owner and backup.
- Auto-generated case summary: what happened, impact, dependencies, decisions needed.
- Recovery plan with accountable actions.
- Executive escalation only for genuine exceptions.
- Root-cause closure required for repeat/high-severity events.

8.14 Post-Handover / Warranty / Service

Extend the relationship beyond keys.

- Move-in support tasks and utility/FM onboarding.
- Defect liability and warranty case management.
- Digital Home Passport with equipment, serial numbers, manuals and warranties.
- Service history retained against the unit.
- Customer satisfaction checkpoints after 7/30/90 days and DLP closure.
- Referral/testimonial/advocacy workflow after positive outcomes.

9. Handover Gate Design

Gate dimension	Illustrative mandatory conditions	Override?
Financial	Required consideration received; TDS verified; approved waivers posted; no unapproved dues.	Authority-controlled only
Legal	Executed agreement / required legal approvals complete.	Normally no
Registration	Registration complete or policy-approved exception.	Authority-controlled
Physical	Construction readiness threshold met; utilities available.	No for safety-critical items
Quality	QA approved; zero critical snags; minor snags within policy.	Limited
Commitments	Critical customer commitments closed or explicitly accepted.	Authority-controlled
Customer	Appointment, identity, nominees/representatives and orientation ready.	Operational
FM / Community	Access, meters, keys, manuals, emergency contacts and onboarding ready.	Operational

10. AI & Intelligence Layer

AI should sit on top of disciplined workflow and data. Do not begin with a generic chatbot. Build the intelligence services first, then expose them through copilots and automated actions.

Engine	Purpose	Example output
Journey Risk Engine	Predict milestone delay and identify drivers.	"Handover 19 Aug: 42% confidence; registration + QA are critical path."
Next Best Action Engine	Prioritize the most valuable action now.	"Secure SRO slot today; improves handover probability by 18 points."
Collection Risk Engine	Predict cash realization and causes of slippage.	"₹48L likely delayed due to bank documentation; customer default risk low."
Commitment Risk Engine	Predict broken promises before due date.	"Customer-facing commitment likely to miss by 4-6 days."
Sentiment / Escalation Engine	Detect deteriorating relationship before formal complaint.	"Negative trend across 3 interactions; proactive RM call recommended."
Document Intelligence	Classify, extract and validate documents.	"PAN valid; name match; address differs from booking record."
Quality Root-Cause Engine	Find recurring defects and operational patterns.	"Waterproofing repeat rate concentrated with contractor X."
Profitability Leakage Engine	Connect service failures to cost/margin.	"₹12.4L quarterly leakage from rework + concessions."

10.1 Copilot Experiences

- Management: "What requires my attention today and why?"
- CRM: "Which customers should I contact today?"
- Collections: "Which ₹2 Cr is most likely to close this week?"
- Projects: "Which units are blocking next month's handovers?"
- Legal: "Which agreements are beyond SLA and what is the blocker?"
- QA: "Which repeat defects need contractor-level intervention?"
- Customer service: "Summarize this customer's history before I call."

11. Customer Experience Layer: My Pranava Home

The customer must not see the internal operating system. They should see a calm, premium, contextual experience that answers what matters without exposing irrelevant internal complexity.

Customer area	Experience
Journey	Clear lifecycle timeline, achieved milestones, next milestone and expected date.
My Home	Unit details, approved specifications, progress imagery and important updates.

Customer area	Experience
Payments	Demand schedule, receipts, outstanding, TDS guidance and secure payment links.
Documents	Required, received, approved and downloadable customer documents.
Registration	Readiness status, document checklist, appointment details and final documents.
Handover	Readiness, appointment, orientation, checklists and signed handover pack.
Requests	Single place to raise questions/service requests and submit/approve unit customisations, quotations, drawings and status.
Commitments	Customer-visible commitments and current status.
Home Passport	Manuals, warranties, product details and service history after handover.

Customer visibility rules: show commitments, milestones, actions required from customer, approved dates and final evidence. Do not show internal blame, employee performance, vendor disputes, internal notes or unapproved forecasts.

12. Moments That Matter: Experiential Journey

Moment	Designed experience
Booking + 24 hours	Premium welcome, relationship manager introduction, journey map, payment/doc checklist and communication preference capture.
Construction milestone	Curated progress update with project-specific photo/video and what happens next.
75-80% completion	Home readiness preview, finance/registration readiness check and personal concierge introduction.
Pre-handover	Guided preparation, document/finance closure plan, snag/QA visibility at appropriate level.
Handover day	Appointment-led homecoming experience, guided walkthrough, digital home passport, keys, utilities/FM orientation, family moment.
7/30/90 days	Structured check-in, service support and satisfaction capture.
Positive closure	Referral, testimonial and advocacy request at the right emotional moment.

13. Role-Based Workspaces

Every role workspace must open in the user's assigned Project scope by default, with a visible Project selector when the user is authorized for multiple Projects. A person may belong to Project-dedicated and shared teams simultaneously; task ownership and escalations must resolve through the applicable Project Team Assignment.

Stakeholder	HomeFlow must answer
Customer	What is happening with my home and what do I need to do?
CRM / Relationship Manager	Who needs attention today, why, and what should I do next?
Sales	Which available units best fit this prospect, which customisation gates are open/closing, and when do those windows expire?
Accounts / Collections	What is due, truly at risk, disputed or loan-dependent?
Loan Coordinator	Which future demands are at risk due to disbursement readiness?
Legal	Which cases require legal judgment, approval or deviation handling?
Registration	Which customers can register, when, and what is blocking them?
Site Engineer	Which unit-level tasks and evidence must be completed?
QA	What needs inspection, reinspection or systemic intervention?
Vendor / Contractor	What defects/actions belong to me, with evidence and SLA?
Handover Team	Which units are truly eligible and which blockers must be cleared?
Department Head	Where is my team missing SLA, creating rework or breaking commitments?
Management	Where must I intervene to protect customers, cash, schedule, reputation and margin?
Design / Technical / Customisation Team	Which change requests need feasibility, costing, drawing release, approval or closure, and what is their schedule/margin impact?

Stakeholder	HomeFlow must answer
Project / Construction	What physical progress changed today, which units/gates are affected, and which holds or exceptions require my decision?
Design / Engineering	Which requested changes require feasibility judgment and which technical rules/gates must be maintained?
Procurement	Which material/order events will close or restrict unit changeability and which approved variations require procurement action?

14. Executive Control Tower

Design principle

Management should see five problems that need intervention, not fifty charts.

- Portfolio: active customers, bookings, units in pre-handover, next 30/60/90-day handovers.
- Cash: due, overdue, true collection risk, forecast, disputes and loan-dependent exposure.
- Project Cash Flow: current-month actual, next-month forecast, 90-day forecast, prior-period actual, forecast variance and confidence by Project with portfolio roll-up.
- Project Performance: customer health, handover risk, collections, SLA, customisation impact, rework/leakage and team workload shown with drill-down to Unit/Booking.
- Experience: Customer Health distribution, sentiment deterioration, commitment breaches, NPS/CSAT.
- Execution: SLA compliance, critical path delays, handovers at risk, critical snags, registration slippage.
- Profitability: rework cost, concessions/waivers, compensation, cost-to-serve, leakage by project/root cause.
- Five management interventions: system-generated, ranked, with owner, rupee/customer impact and decision required.

15. Profitability and Margin Protection

Customer experience must be excellent and economically disciplined. HomeFlow should explicitly measure the cost of failures, exceptions and concessions.

Economic object	Examples
Revenue / receivable	Sale value, upgrades, services, scheduled collections, realized collections.
Commercial leakage	Discounts, free items, unapproved promises, waivers, interest reversals.
Service leakage	Compensation, hospitality, special transport, repeated manual interventions.
Quality cost	Rework labor/material, vendor back-charges, repeat snag cost.

Economic object	Examples
Delay cost	Extended staffing, holding cost, interest implications, customer compensation.
Cost-to-serve	CRM effort, visits, escalations, legal effort, exceptional processing.
Unit / booking contribution	Original planned margin vs current margin after leakage.
Customisation economics	Variation selling price, incremental vendor/internal cost, taxes, discounts/waivers, abortive work, contribution and margin.

16. SLA, Escalation and Accountability Framework

Level	Trigger	Response
L0	Normal action within SLA	Owner executes; dashboard only.
L1	Approaching SLA / forecast risk	Owner + backup notified; recommended recovery action.
L2	SLA breached / customer milestone impacted	Department manager owns recovery plan.
L3	Repeated breach / material financial or customer impact	Functional head intervention.
L4	Critical customer/reputation/legal/safety impact	CX/COO/management alert with decision pack.

Every escalation should include a short system-generated decision pack: what happened, current impact, critical dependencies, actions already taken, owner, next deadline, recommended decision and evidence links.

- All plan and SLA analytics must roll up by Project, stage, department, role, owner, unit and customer/booking.
- SLA clocks and plan dates are distinct. SLA may pause only for configured reasons; plan/forecast dates continue to support milestone prediction and management variance analysis.
- System statuses must include On Track, Due Soon, At Risk, Overdue, Completed On Time and Completed Late, derived from plan/forecast/actual rather than manually painted status.
- Changing a planned date must not erase history. Store baseline, revisions, actor, timestamp, reason and approval where policy requires it.
- Planned duration may be in working days or calendar days and may be inherited from a Pranava default, Project override, process/stage override or exceptional record-level override.
- Each configurable item may store: planned duration, baseline planned start/end, current planned start/end, forecast start/end, actual start/end, owner, dependency, gate, customer visibility, calendar type, pause reason, variance and delay reason.

HomeFlow shall use a Universal Timeline & SLA Engine across journey stages, sub-stages, tasks, approvals, collections, legal, registration, change requests, QA/snags, handover and post-handover service.

17. Controlled Openness and Permissions

Principle	Design requirement
Need-to-know	Users see information required to perform their role.
Customer transparency	Customers see approved milestones, responsibilities, commitments and evidence relevant to them.
Internal confidentiality	Internal notes, staff performance, vendor disputes and unapproved forecasts remain internal.
Financial sensitivity	Discounts, margins, concessions and cost-to-serve restricted by role.
Immutable audit	Changes to commercial terms, commitments, approvals and closures remain traceable.
Explicit overrides	Hard-gate overrides require named authority, reason and evidence.

18. Integration Blueprint

Integration architecture is optional and system-agnostic. HomeFlow is the operating system of record for the workflows defined in this specification unless a future governance decision explicitly designates another authoritative source for a specific data domain. Any later connector must use generic mappings/adapters and must not introduce vendor-specific assumptions into the core Project, Unit, Booking, Customer or workflow models.

System / channel	HomeFlow relationship
Sales CRM (e.g., Tranquil)	Booking/customer source, sales handover, source/channel history.
Optional Construction / Project System	Optional future connector only. HomeFlow must natively support Project/Unit progress, component status, evidence, readiness and changeability inputs. If a construction/project platform is adopted later, generic APIs/events may import approved milestones or activity states and map them to the Unit Digital Twin without changing the core data model.
ERP / Accounting	Demand, receipt, ledger, TDS, waivers, credit notes, financial clearance.
Bank / payment gateway	Payment status and reconciliation where APIs are available.
Loan partners / coordinators	Sanction/disbursement status; document readiness.
DMS / storage	Authoritative document files, versions and retention.
WhatsApp / Email / SMS	Approved outbound communication and interaction logging.

System / channel	HomeFlow relationship
eSign / eStamp / registration services	Execution workflow and evidence where technically/legal available.
Facility Management	Post-handover service and unit asset history.
BI / Data warehouse	Portfolio analytics, historic benchmarking and model training.

Integration rule: external systems are optional. When one is configured, each integrated data domain must have a defined system-of-record owner and reconciliation rule. HomeFlow must not silently overwrite authoritative facts, and it must continue to support the same business process when the external connector is unavailable or absent.

19. Reporting and KPI Framework

All reporting must support the hierarchy Portfolio -> Project -> Phase/Tower/Block -> Unit and the organizational view Portfolio -> Project Team / Shared Team -> Role -> Owner. Project is a mandatory analytical dimension for finance, CX, execution, quality and profitability.

- Collections Actual vs Forecast by Project and period.
- Next-month / 30-60-90 day cash realization forecast with confidence bands.
- Forecast variance and forecast accuracy by Project, team, reason code and ageing bucket.
- Project-level receivables waterfall: opening outstanding, fresh demands, expected receipts, recoveries, waivers/adjustments and closing outstanding.
- Project team productivity and SLA compliance, while preserving shared-team attribution where the same team serves multiple Projects.

Domain	Core KPIs
Customer experience	NPS, CSAT, response SLA, sentiment trend, escalation rate, referral rate.
Commitments	On-time commitment %, pre-breach recovery %, broken promises by root cause.
Finance	Collection efficiency, overdue %, true-risk amount, forecast accuracy, promise-to-pay conversion.
Legal / registration	Agreement TAT, deviation TAT, registration readiness, registration predictability.
Quality	First-pass QA, critical snag rate, repeat snag rate, snag closure TAT, rework cost.
Handover	Handover predictability, on-time %, first-time-right %, hard-gate overrides.
Operations	SLA compliance, action ageing, reassignments, dependency delay, queue health.

Domain	Core KPIs
Profitability	Concession leakage, rework, delay cost, compensation, cost-to-serve, margin erosion.
Customer changes / customisations	Request-to-feasibility TAT, quote TAT, approval conversion, on-time execution, post-freeze change %, variation margin, rework due to wrong revision, change-driven handover delay.
Changeability / personalisation	% available units with fresh gate data, gate forecast accuracy, requirement-to-unit match conversion, change-window hold conversion, post-gate exception rate, change margin and schedule impact.

- Project Journey health: active journeys by On Track / At Risk / Overdue and top bottleneck stages.
- Stage/SLA predictability: median and percentile cycle time, on-time %, forecast accuracy, pause reasons and repeat delay causes.
- Journey plan-vs-forecast-vs-actual variance by Project, stage, department and owner.

20. UX / Screen Architecture

Primary screen	Purpose
Home / My Day	Personal action cockpit, upcoming risks, approvals and customer contacts.
Customer 360	Relationship twin: bookings, financials, docs, communication, commitments, experience.
Unit 360	Property twin: specs, construction, QA, snags, evidence, assets, warranty.
Booking 360	Commercial/lifecycle bridge and readiness convergence view.
Portfolio Control Tower	Project/portfolio risk, cash, handover, experience and profitability.
Functional Queues	Collections, legal, registration, QA, snag, handover, documents.
Journey Timeline	Milestones, critical path, blockers, events and next forecast.
Customer Portal / App	Approved customer-facing journey and home experience.

Primary screen	Purpose
Admin / Policy Studio	Workflow templates, SLAs, gates, approval matrix, roles and notifications.
Unit Changes & Customisations	Request intake, feasibility, quotation, approvals, payment, drawing/spec revision, execution, QA, customer acceptance and as-built closure.
Unit Progress & Changeability Control	Project-owned bulk/unit update interface; component progress, gate transitions, source/freshness and impact preview.
Sales Inventory Changeability View	Available-unit comparison with construction state, changeability score, open/closing gates, expiry forecast and requirement-fit filters.
Personalisation Discovery & Unit Match	Capture Must Have/Preferred personalisation needs and rank units by live requirement compatibility.

- Project Journey Control: portfolio/project view of active journeys, bottleneck stages, upcoming milestones and at-risk/overdue items.
- Journey Timeline: Baseline / Current Plan / Forecast / Actual toggles with variance explanation and critical blockers.
- Project Journey Designer: visual template showing stages, parallel streams, dependencies, gates, planned duration and customer-visible milestones.

21. Admin / Policy Studio

- Project-level workflow templates and product variations (apartment, villa, office, plotted development).
- Conditional task rules based on customer type, loan status, NRI status, project, unit, commercial scheme and exceptions.
- SLA policies and calendars including holidays and pause reasons.
- Approval authority matrix by amount, deviation type, project and role.
- Handover gate configuration with hard/soft conditions.
- Communication templates, approval and version control.
- Score weights, thresholds and model governance.
- Role/permission matrix and field-level sensitivity.
- Escalation routing and management thresholds.
- Customisation policy by project/product: allowed change catalogue, freeze dates, technical constraints, quotation validity, payment gates and cancellation rules.
- Variation approval matrix by change type, value, margin, schedule impact and post-freeze exception severity.
- Change log and effective-dated policy versions.
- Change-gate rule studio by project/product/unit type/component with trigger event, state transition, hard/soft classification and customer-facing explanation.
- Gate-expiry forecast source mapping: construction schedule, procurement milestone, manual approved date or integration event.

- Change Window Hold policy: eligible gates, maximum duration, approvers, blackout activities, concurrent-hold limits and automatic expiry.
- Data freshness thresholds and stale-state warnings for Sales/CRM when project updates are older than policy.
- Project master and hierarchy configuration: Phase, Tower/Block, Floor/Cluster and Unit inheritance rules.
- Project Team Assignment matrix: dedicated vs shared team, departments/roles, effective dates, primary/backup owner, capacity, escalation manager and permitted Projects.
- Forecast policy studio: forecast horizons, probability rules, source precedence, reschedule treatment, overdue recovery assumptions, confidence thresholds and scenario definitions.
- Period calendar: financial month, week, quarter, custom periods and project-specific milestone calendars.
- Project-specific cash-flow targets and collection plans with portfolio aggregation.
- Template versioning: effective date, version, approver, impacted active journeys, migration rule and audit trail. Existing journeys must not be silently reshaped when a template changes.
- Parallel stream configuration: allow Finance, Construction, Legal/Documentation, Customisation, Commitments and Communication streams to run concurrently and converge through milestone gates.
- Stage-level customer visibility: define what the customer sees, what wording is used, and whether forecast dates are shown as date, window or milestone.
- Timeline policy: planned duration, working/calendar day basis, start trigger, baseline/current plan, forecast rules, actual capture and variance tolerance.
- Project Journey Template inheritance: copy/inherit the Pranava Standard, then allow controlled Project-specific overrides without changing code.
- Pranava Standard Journey Template: enterprise default stages, sub-stages, tasks, gates, dependencies and customer-visible milestone labels.

22. Data Quality, Audit and Governance

- Single master identifiers for Project, Unit, Booking, Customer and Applicant.
- Project is immutable on a Unit after controlled master creation except through audited master-data correction. Project on downstream operational records must derive from the Unit/Booking relationship.
- Forecast snapshots must be versioned by as-of date so management can compare what was expected at month start, subsequent revisions and actual realization.
- Deduplication and merge process for customers without losing history.
- Effective dating for ownership changes, commercial revisions and policy changes.
- No deletion of material financial/legal/commitment history; use superseded/cancelled states.
- Mandatory structured reason codes for returns, delays, overrides, waivers, cancellations and escalations.
- Immutable event/audit log for consequential changes.
- Role-based export control and sensitive-field masking.
- Immutable linkage from original unit specification to every approved change and final as-built revision.
- No deletion of superseded drawings, rejected change requests or commercial variation history; retain status and effective dates.
- AI output logging: model/version, confidence, user action and eventual outcome for continuous learning.

23. Notifications and Communication Rules

Avoid notification fatigue. The system should prioritize digestible actions and exceptions rather than sending an alert for every state change.

Type	Rule
Employee action	In-app by default; email/WhatsApp only for urgent or opted-in categories.
Daily digest	Prioritized My Day with overdue, due today, at-risk and approvals.
Customer update	Event-driven and value-adding; avoid repetitive status noise.
Pre-breach alert	Before commitment/SLA/milestone failure when recovery is still possible.
Management alert	Only material exception thresholds; include concise decision pack.
Quiet hours / preferences	Respect customer and employee communication windows and channel preferences.

24. Priority Roadmap

P0 architecture: enforce Project hierarchy across all core objects, introduce Project Team Assignment, implement Project-wise collections actual/forecast/variance, and establish the controlled Legal Document Factory foundation before adding advanced AI cash forecasting or autonomous document agents.

Priority	Capability	Why now	Indicative outcome
P0 - 0-90 days	Canonical data model: Project/Unit/Booking/Customer + universal Action object	Foundation for every later intelligence layer.	Removes fragmented records and duplicate logic.
P0 - 0-90 days	Separate Unit Readiness, Customer Readiness and Handover Readiness	Current percentages can be ambiguous.	Clear, explainable operational control.
P0 - 0-90 days	Hard handover gates + blocker/dependency model	Protects customer experience and operational integrity.	Fewer preventable bad handovers.
P0 - 0-90 days	Promise Ledger + pre-breach rules	Broken promises are a major trust leak.	Higher commitment reliability.
P0 - 0-90 days	My Day + universal action prioritization	Immediate employee productivity gain.	Less chasing; clearer accountability.
P0 - 0-90 days	Collections reason codes + true-risk view	Outstanding alone is not risk.	Better cash focus and forecast.
P1 - 3-6 months	Customer 360 + Unit 360 + Booking 360 redesigned around digital twins	Creates one truth for all stakeholders.	Faster decisions and fewer handoffs.
P1 - 3-6 months	Evidence-based QA / snag	Makes readiness	Lower rework and repeat

Priority	Capability	Why now	Indicative outcome
	closure	trustworthy.	defects.
P1 - 3-6 months	Customer portal "My Pranava Home"	Creates experiential transparency.	Fewer status calls; higher confidence.
P1 - 3-6 months	Document intelligence	Cuts manual review and errors.	Faster legal/registration processing.
P1 - 3-6 months	Predictive journey / handover / collection risk	Moves from reporting to prevention.	Earlier intervention and better forecasts.
P2 - 6-12 months	AI copilots by role	Useful only after data/workflow foundation.	Natural-language access and action preparation.
P2 - 6-12 months	Digital Home Passport + post-handover warranty	Extends lifecycle and brand relationship.	Better service and long-term asset history.
P2 - 6-12 months	Profitability leakage + cost-to-serve analytics	Connects CX to economics.	Margin protection and root-cause investment decisions.
P2 - 6-12 months	Vendor/contractor performance learning	Closes loop from post-sales to construction.	Lower defects and stronger vendor accountability.
P0 - 0-90 days	Customer Change Request / Unit Customisation workflow + change-freeze + approval/payment controls	Customer changes can create hidden schedule and margin risk if handled informally.	Controlled flexibility, fewer disputes, protected margin and clear site instructions.
P1 - 3-6 months	Variation catalogue, drawing/version integration, as-built Unit Twin and profitability analytics	Turns one-off change handling into a scalable product capability.	Faster quotations, safer execution, better contribution and permanent unit history.
P0 - 0-90 days	Unit Progress + Changeability Engine v1	Sales decisions require live physical truth before and after booking.	Project-owned progress states automatically derive open/closed/conditional customisation gates.
P1 - 3-6 months	Sales requirement-to-unit matching + gate expiry forecast	Turn personalisation into a controlled sales advantage.	Higher-fit unit selection, fewer invalid promises, improved conversion.
P1 - 3-6 months	Controlled Change Window Hold	Protect legitimate sales opportunities without uncontrolled construction disruption.	Time-boxed, approved holds with automatic expiry and audit trail.

Priority	Capability	Why now	Indicative outcome
P0 - 0-90 days	Configurable Project Journey Template + Universal Timeline/SLA Engine	Avoids hardcoded project SOPs and makes plan/forecast/actual measurable across all workflows.	One generic operating model with controlled Project-specific variation and reliable timeline analytics.

25. 90-Day Implementation Plan

Within the foundation phase, deliver the Project dimension migration, project/team assignment model, Project selector/filter behavior, collections forecast snapshots, Actual vs Forecast dashboards, drill-down from Project cash flow to Booking/Unit, and the first production-ready document templates with governed merge fields, validation, approvals and version history.

Sprint	Primary deliverables
Weeks 1-2	Freeze terminology, canonical IDs, Customer/Unit/Booking schemas, lifecycle stages, action states, ownership rules. Define component taxonomy and unit-level physical-state ownership. Define Pranava Standard Journey Template, Project inheritance/override rules, universal date model (baseline/current/forecast/actual), SLA calendars and trigger logic.
Weeks 3-4	Implement readiness score separation, hard/soft gates, dependency model and blocker explanation. Define change-request taxonomy, freeze rules, feasibility/impact assessment and approval matrix. Build Unit Progress Control v1 and rule-derived changeability states with hard/soft/conditional classifications.
Weeks 5-6	Universal Action object, My Day cockpit and departmental queue normalization.
Weeks 7-8	Promise Ledger, pre-breach alerts, escalation decision packs and reason-code taxonomy. Implement customisation quotation/customer approval/payment-release workflow and variation audit trail. Expose live changeability to Sales/CRM inventory and implement stale-data warnings.
Weeks 9-10	Collections true-risk model, promise-to-pay, loan dependency and management cash view.
Weeks 11-12	Executive control tower v1, KPI baselines, audit

Sprint	Primary deliverables
	checks, pilot project rollout and user acceptance. Pilot controlled unit changes on one project and baseline variation margin/TAT metrics. Pilot requirement-to-unit matching and gate-expiry forecasts on one project/unit type.

26. Acceptance Criteria for HomeFlow 2.0 Foundation

- Every active booking resolves to exactly one current unit and one or more valid applicants/customers.
- Unit history remains intact when booking/customer changes.
- Every actionable record appears in the universal action engine with owner, SLA and evidence requirement.
- Every readiness score is explainable down to component/blocker level.
- Hard handover gates cannot be bypassed without configured authority and audit reason.
- Every customer-facing commitment has owner, due date, status, dependencies and evidence.
- Every overdue collection has a structured reason and next action.
- Management can identify the top five portfolio interventions without navigating multiple reports.
- Every financial and operational record can be traced Portfolio -> Project -> Unit -> Booking/Customer without duplicate manual project tagging.
- A user assigned to one Project sees that Project by default; a shared-team user can switch only among authorized Projects and retains correct task/escalation context.
- Management can view last-period actual collections, current-period actual-to-date, next-month forecast and 30/60/90-day forecast by Project and portfolio.
- Every forecast value can be drilled to contributing Booking/Unit records and shows source, expected date, probability/confidence and last update.
- The system retains forecast snapshots and calculates forecast-to-actual variance without overwriting prior forecasts.
- Customer-facing information never exposes internal notes or unapproved assumptions.
- Critical workflows and edits have a complete audit trail.
- Every customer-requested unit change has a unique request ID, structured scope, impact assessment, approval state and auditable disposition.
- No change can be released to Site before configured feasibility, commercial approval, customer acceptance and payment gates are satisfied.
- Site, QA and Procurement can identify the current released drawing/specification revision; superseded revisions are visibly locked.
- Every completed customisation updates the permanent Unit Digital Twin / as-built record and preserves variation economics.
- Every saleable unit has a current physical-progress record and an explainable changeability state by configured change category.
- Sales and CRM can view but cannot directly edit physical progress or technical gate states.
- When a configured construction/procurement event changes, affected gates are automatically re-evaluated and the event is audited.
- Sales can filter/compare available units by changeability and see source timestamp/freshness before discussing personalisation with a prospect.

- A prospect's Must Have / Preferred personalisation requirements can be compared with available units without implying technical approval.
- Closing gates display an expected expiry date/event; stale forecasts are clearly identified.
- Any Change Window Hold is time-bound, Project-approved when construction is affected, automatically expires, and leaves a complete audit trail.
- Hard Closed gates cannot be reopened by Sales, CRM or ordinary management override.
- Unit-level exceptions remain possible when actual unit progress differs from tower/floor bulk progress, with reason and audit.
- Approved users can generate an AOS, Sale Deed or Lease Agreement only from an approved template version valid for the relevant Project/transaction context.
- Generated documents auto-populate authoritative Project, Unit, Booking, Customer and financial fields and prevent release when mandatory source data is missing or inconsistent.
- Every generated document records template version, data snapshot, clause selections, overrides/deviations, creator, reviewers, approvals, timestamps and final executed artifact.
- Any manual change to a protected commercial/legal field requires configured authority and an auditable reason; ordinary users cannot silently edit generated legal text or values.
- Legal can compare revisions and identify substantive clause/value changes before approval; superseded drafts remain accessible but cannot be mistaken for the current executable version.
- Final executed/registered documents are locked, checksum/version identified, and visible from Project, Unit, Booking and Customer records according to permissions.

27. What Not to Build Yet

- A generic chatbot before the underlying data, rules and actions are trustworthy.
- Dozens of scores that nobody can explain or act upon.
- Another standalone chat stream that duplicates comments, email and WhatsApp.
- Manual progress percentages where evidence can derive the state.
- Executive dashboards overloaded with charts instead of ranked exceptions.
- AI auto-sending consequential customer communication without approval controls.
- Duplicated accounting/construction master data when authoritative systems already exist.
- Custom features that solve one project exception but weaken the common operating model.

28. North-Star Operating Rhythm

Cadence	Operating rhythm
Daily employee	My Day: execute prioritized actions, resolve blockers, capture evidence and commitments.
Daily department head	Queue health, SLA risks, blocked items, capacity and recovery plans.
Daily management	Five interventions: customer, cash, handover, reputation, profitability.
Weekly project / CX review	Upcoming 30/60-day handovers, customer health, collection risk, repeat defects, broken commitments.
Monthly operating review	KPI trend, root causes, vendor/team performance, policy changes and margin leakage.

Cadence	Operating rhythm
Quarterly product review	Workflow simplification, model accuracy, automation opportunities and customer feedback.

29. Final Target State

The HomeFlow 2.0 end state

A customer opens My Pranava Home and immediately understands their journey. An employee opens My Day and immediately knows what matters. A project or functional head sees where the process is failing. Management sees only material exceptions. The unit retains a permanent digital history. AI predicts slippage before it becomes a complaint. And every operational failure can be traced to its customer, schedule and margin impact.

The five tests for every future feature are simple: Does it improve customer trust? Does it eliminate chasing? Does it expose accountability? Does it predict problems earlier? Does it protect or improve margin? If the answer to none is yes, it should not be built. For unit customisations, add a sixth test: Does this change preserve the permanent as-built truth of the unit while protecting schedule and margin?

30. Emergent Build Instructions - Unit Progress, Configuration & Changeability

This section is implementation guidance for the Emergent product team. Build this capability as shared platform infrastructure used by Sales inventory, CRM, Customer Change Requests, Unit 360, readiness and management reporting. Do not build it as a disconnected customisation screen.

30.1 Required core objects

Object	Minimum fields / relationships
UnitProgressState	unit_id; component_id; state_code; progress_pct if applicable; actual_date; planned_next_event; source_system; source_record_id; updated_by; updated_at; confidence/freshness.
ChangeCategory	category_id; project/product applicability; room/trade/system; customer-facing label; technical owner; default policy.
ChangeGateRule	rule_id; category_id; trigger component/event; condition; resulting state; hard/soft classification; effective dates; exception authority.
UnitChangeGate	unit_id; category_id; current_state; reason_code; source_event; expected_close_at; closing_event;

Object	Minimum fields / relationships
	last_evaluated_at; freshness_status.
ProspectPersonalisationNeed	opportunity/lead_id; requirement; importance (Must Have/Preferred); category mapping; notes.
UnitRequirementMatch	opportunity_id; unit_id; compatibility_score; matched/open items; conditional items; closed items; generated_at.
ChangeWindowHold	hold_id; opportunity_id/customer_id; unit_id; category/gate; requested_from/to; reason; project impact; approver; status; expires_at; release_reason.

30.2 Required screens and UX

- Project > Unit Progress Control: hierarchy filters, bulk update, unit exception, affected-gate preview, save with audit reason where required.
- Unit 360 > Changeability: visual matrix by room/trade/change category with current state, reason, expected closure event/date, source and last update.
- Sales > Inventory: add changeability score, quick gate chips, closing-soon indicator, filters and side-by-side comparison.
- Sales > Personalisation Discovery: capture Must Have/Preferred needs and return ranked available units with compatibility explanation.
- CRM > Customer Change Request: show live gate state at request creation; never block request capture solely because a gate is closed; route to correct feasibility/exception decision.
- Project > Change Window Holds: pending holds, planned activity conflict, approve/reject, expiry and release.
- Management > Exceptions: stale gate data, high-value exception requests, holds affecting schedule, post-gate changes, change margin/schedule impact.

30.3 Rule execution requirements

- Re-evaluate relevant UnitChangeGate records whenever UnitProgressState, planned activity date, procurement status, approved variation or policy rule changes.
- Use event-driven recalculation where integrations support it; otherwise scheduled reconciliation plus stale-state warnings.
- Do not derive technical feasibility from overall construction percentage alone. Use mapped component/trade events.
- If data is stale beyond configured threshold, Sales/CRM must see "Verification Required" rather than a falsely precise open/closed promise.
- For a gate moving from Open to Conditional/Exception/Hard Closed, log previous state, new state, source event and affected active prospects/change requests.
- If an active approved Change Window Hold conflicts with an incoming project progress event, warn Project before commit and require explicit resolution.
- No API or UI path available to Sales/CRM may directly mutate UnitProgressState, ChangeGateRule or technical hard-gate state.

30.4 Key workflow logic

Scenario	System behaviour
Unsold early-stage unit	Maintain progress/gates independent of customer. Sales sees high customisation flexibility and can match prospect needs.
Prospect requests feature before booking	Show live gate + fit; optionally request time-boxed hold. Do not create site instruction.
Customer books unit	Booking links Customer Digital Twin to existing Unit Digital Twin; existing gate state continues unchanged.
Customer raises late request	Capture request; evaluate current gate; route normal/conditional/exception/reject based on feasibility and hard-gate policy.
Project updates progress in bulk	Preview affected units/gates; commit state; re-evaluate gates; notify only affected active workflows.
Unit differs from tower state	Authorized unit exception overrides bulk state for that unit, with reason/source/evidence.
Gate closes while quote pending	Re-evaluate feasibility and require quote/version refresh if impact changes; do not silently execute obsolete commercial/technical assumptions.

30.5 Minimum acceptance tests for Emergent

- Given two units in the same project with different physical progress, the same change category can correctly show Open for one and Exception Only for the other.
- When Project marks electrical first-fix as commenced for a unit, mapped electrical gates change according to configured rule without Sales editing anything.
- When a tower-level bulk update affects 40 units, the UI previews affected units/gates and allows an authorized unit-level exception.
- Sales can filter inventory to units where Kitchen Layout and Electrical Changes are Open and compare at least three units side by side.
- A Closing gate displays a predicted closure date/event and becomes stale/verification-required when underlying schedule data exceeds freshness policy.
- A Must Have requirement that is Hard Closed materially reduces the Unit Requirement Match score and is explicitly explained.
- A customer change request can still be created after Registration/Handover scheduling; the system routes it to exception or post-handover treatment instead of deleting the request path.
- An approved Change Window Hold expires automatically and can no longer block project progress after expiry.
- Every gate transition and override contains timestamp, actor/source, previous state, new state and reason/event.
- Hard Closed technical/statutory gates cannot be reopened by ordinary override APIs or UI permissions.

31. Emergent Build Instructions - Project Architecture & Collections Forecasting

Build Project as a platform-level partition and analytical dimension, not as an optional filter bolted onto reports. All new features must work at Unit/Booking detail and roll up reliably to Project and portfolio. Collections forecasting must be explainable, versioned and drillable.

31.1 Required hierarchy and ownership model

- Canonical hierarchy: Portfolio -> Project -> Phase/Tower/Block/Cluster -> Floor (where applicable) -> Unit -> Booking -> Customer/Applicant.
- Required entities: Project, ProjectHierarchyNode, Unit, Booking, Team, ProjectTeamAssignment, UserRoleAssignment, ForecastSnapshot, CollectionForecastLine, ForecastScenario and ForecastAssumption.
- ProjectTeamAssignment fields: project_id, team_id, department, role_scope, assignment_type [Dedicated/Shared/Central], primary_owner, backup_owner, effective_from, effective_to, capacity/weight, escalation_manager, permissions.
- Every downstream record must carry a derived project_id for query performance and row-level security, but the system must validate that it matches the linked Unit/Booking source of truth.
- Support one team serving multiple Projects and one Project using multiple specialist/shared teams. Do not hard-code one CRM team = one Project.

31.2 Collections forecasting model

- Create a CollectionForecastLine per expected receipt event, linked to Project, Unit, Booking, Customer, demand/milestone, owner/team, expected_date, gross_amount, expected_amount, probability, confidence, source_type, reason_code and last_evidence_date.
- Supported source types at minimum: Contractual Due, Overdue Recovery, Promise-to-Pay, Loan Disbursement, Registration/Final Demand, Approved Reschedule, Manual Finance Override and Scenario-only Future Sales. Scenario-only Future Sales must be visually and mathematically separated from booked receivable forecast.
- Forecast periods must be configurable. Default quick views: Previous Month Actual, Current Month Plan vs Actual-to-Date, Next Month Forecast, Next 90 Days, Quarter and Custom Range.
- Store immutable forecast snapshots at configured checkpoints (for example month-start and weekly) and allow revised forecasts as new versions. Never overwrite history.
- For each Project calculate: opening outstanding, demands due in period, overdue at start, expected contractual collection, expected overdue recovery, expected loan disbursement, expected rescheduled amount, total expected receipts, actual receipts, forecast variance, closing expected outstanding and confidence.
- Probability weighting must be rule-based first and explainable. Example inputs: ageing, historical payment behavior, promise-to-pay quality, bank disbursement stage, missing documents, dispute status, milestone readiness and recent customer interaction. AI can refine probability later but cannot hide the drivers.
- Forecast edits require reason, actor and evidence. Finance/Collections own forecast overrides; CRM may update customer promise signals but cannot silently change finance-owned forecast amounts.

31.3 Required screens and UX

- Executive / Finance > Project Cash Flow Planner: Project cards plus portfolio total, period selector, Actual vs Forecast vs Revised Forecast, confidence, variance, and receivables waterfall.

- Project > Collections Forecast: timeline/table of forecast lines with Unit, Customer, expected date, due amount, expected amount, source, probability, owner, blocker and next action.
- Portfolio > Project Comparison: compare projects on current-month collections, next-month forecast, overdue, forecast accuracy, collection efficiency and risk.
- Project 360 header: common project context bar showing project name, phase/tower filters, active units, bookings, receivables, next-30-day handovers, customer health and assigned teams.
- Every major module screen must include Project context. If opened from a Project 360 page, retain that Project context when navigating Collections, CRM, Readiness, Snags, Handover, Commitments, Escalations and Reports.
- Sales/CRM/Project/Finance workspaces must show Assigned Project(s) and allow authorized users to switch between projects without losing role-specific filters.

31.4 Forecast calculations and behavior

- Actual collection is based on authoritative posted/reconciled receipt date and amount from finance system. Do not count unverified payment promises as actual cash.
- Current-month plan should show Plan, Forecast-at-Month-Start, Latest Forecast, Actual-to-Date and Projected Month-End, with variance against each relevant baseline.
- Next-month forecast should include receipts whose expected realization date falls in next month even if the contractual due date is earlier/later, while retaining the contractual date for audit.
- When a demand, receipt, reschedule, loan stage, promise-to-pay or dispute state changes, recalculate affected forecast lines and create a new latest forecast version without mutating historical snapshots.
- Allow drill-down from every Project total to the exact Unit/Booking lines contributing to the number.
- Prevent double counting when one receivable is represented by multiple signals (for example contractual demand plus loan disbursement). Use one canonical forecast line with multiple supporting signals or explicit precedence rules.

31.5 Minimum acceptance tests for Emergent

- A Project-dedicated CRM user logs in and sees only assigned Project work by default; a shared CRM manager can switch among authorized Projects and view a portfolio roll-up.
- A receipt posted against a Booking automatically appears under the correct Project without the user manually selecting Project.
- Finance can select Previous Month and see Project-wise actual collections, then select Next Month and see the forecast with drill-down to Unit/Booking.
- The Project total equals the sum of its forecast lines and the portfolio total equals the sum of authorized Projects, subject to explicit elimination rules.
- Month-start forecast remains unchanged after a mid-month revision; Latest Forecast changes and variance is calculated against both Month-start Forecast and Actual.
- A partially disbursed loan case updates expected receipt timing without double counting the same demand.
- A customer Promise-to-Pay changes forecast confidence/expected date according to configured rules, but does not become Actual until a reconciled receipt is posted.
- Future-sales scenario cash is visible only when scenario mode is enabled and is never mixed into committed post-sales receivable forecast by default.
- Project/team assignments are effective-dated: changing the CRM team next month does not rewrite historic ownership and performance attribution.
- All Project dashboards can drill to Unit and Booking detail, and all Unit/Booking records can navigate back to Project 360.

32. Emergent Build Instructions - Legal Document Factory & Intelligent Document Generation

Build document generation as a governed transaction/document platform, not as free-form mail merge. The system must use approved templates, authoritative HomeFlow data, controlled clause logic, validation, approvals, versioning and immutable execution records. The objective is faster preparation, fewer errors, stronger legal/commercial control and a polished customer experience.

32.1 Supported document families

- At minimum support: Agreement of Sale (AOS), Sale Deed / Conveyance Deed, Lease Agreement, Leave & License where applicable, Addendum / Supplemental Agreement, Allotment/Confirmation Letter, Demand Letter, Payment Receipt/Statement, NOC, Possession/Handover Letter, Customer Declaration/Undertaking, Customisation/Variation Agreement, Cancellation/Transfer documentation and approved project-specific letters.
- Document families must be configurable. Do not hard-code one universal AOS or Sale Deed across all Projects, entity structures, property types or jurisdictions.
- Each template version must define applicable Project(s), developer/legal entity, property type, transaction type, jurisdiction, effective-from/effective-to dates, language, status [Draft/Under Review/Approved/Retired] and approving authority.

32.2 Required core objects

- DocumentTemplate: template_id, document_family, name, project_scope, legal_entity, property_type, transaction_type, jurisdiction, effective dates, version, status, owner, approver, source file and checksum.
- MergeFieldDefinition: field_key, label, source_object, source_path, data_type, mandatory/optional, formatting rule, fallback policy, editable flag, validation rule and sensitivity classification.
- ClauseLibrary: clause_id, clause_type, approved text, applicability rules, mandatory/optional status, precedence, legal owner, effective dates and version.
- ClauseSelectionRule: template_id, clause_id, conditions such as Project/property/customer/payment plan/loan/customisation/jurisdiction, default action and exception approver.
- GeneratedDocument: generated_document_id, template_version, Project, Unit, Booking, Customer/applicants, generation timestamp, data snapshot ID, selected clauses, status, current version, owner and final artifact reference.
- DocumentDeviation: field/clause affected, original approved value/text, proposed value/text, reason, commercial/legal impact, requested_by, approval chain, outcome and timestamp.
- DocumentApproval: reviewer, role, stage, decision, comments, timestamp and evidence. Support parallel or sequential review where configured.
- ExecutionRecord: execution method [wet-sign/eSign/registered], signatories, execution date, registration/SRO reference where relevant, final checksum, final file, stamping details and archive status.

32.3 Data source and auto-population rules

- Project, Unit, Booking and Customer are mandatory context for unit-sale legal documents. Lease documents must similarly resolve to Project/Unit plus lessor/lessee and commercial lease context.
- Populate all trusted fields from authoritative sources: Project master, Unit Digital Twin, Booking/commercial master, applicant/customer master, collections/finance, registration, approved customisations and approved commercial deviations.

- Do not ask users to retype a value that already exists in a trusted source. If correction is required, route the user to correct the source record or create a governed override with approval.
- At generation time, freeze a data snapshot so the exact values used to create Version 1 can always be reconstructed even if master data later changes.
- For joint applicants, companies, trusts, NRIs, POA holders or other special cases, support structured party roles and conditional sections rather than manual paragraph editing.

32.4 Pre-generation validation engine

- Before generation, show a readiness panel: Ready / Warning / Blocked. Block only on configured legal/commercial mandatory failures; allow warnings for reviewable conditions.
- Validate applicant names and roles, PAN/identity availability, address consistency, Project/legal entity, Unit identity/area/specification, consideration, taxes/charges, payment plan, approved discounts/deviations, registration data and other template-specific mandatory fields.
- Cross-check numbers in words vs numeric amounts at render time, totals/subtotals, dates, percentages and schedule references. The generated document must not contain unresolved placeholders.
- Detect conflicting source values and require explicit resolution before release. Example: booking value differs from finance-approved consideration or Unit area differs from approved unit master.
- Validation output must be explainable and link users directly to the source field requiring correction.

32.5 Clause engine and controlled legal intelligence

- Build documents from approved base template + Clause Library + rule-selected schedules/annexures. Do not use unrestricted generative AI to invent legal clauses.
- Conditional clauses may depend on Project, property type, customer type, NRI/POA status, loan status, payment plan, approved customisation, lease terms, jurisdiction, statutory status or approved commercial exception.
- Allow Legal to designate clauses as Locked, Parameterized or Negotiable-with-Approval. Locked clauses cannot be edited by Sales/CRM.
- Where a negotiated deviation is proposed, create a DocumentDeviation and route it through Legal/Commercial approval before the changed text can enter an executable version.
- AI may summarize differences, suggest a known approved clause, identify missing clauses or flag anomalies, but final legal wording must come from an approved library/template or explicit Legal-approved deviation.

32.6 Generation, review and execution workflow

- Canonical workflow: Select Document -> Readiness Check -> Generate Draft -> Automated Validation -> Internal Review -> Legal/Commercial Approval as applicable -> Customer Review -> Approved-for-Execution -> eSign/Wet Sign/Registration -> Final Executed Copy -> Archive.
- Support draft watermark/status labeling and prevent customers or staff from mistaking Draft/For Review versions for executable documents.
- Every revision creates a new immutable version. Never overwrite a previously reviewed or customer-shared version.
- Provide side-by-side/redline comparison with structured change summary: financial values changed, party details changed, clause changed, schedule/annexure changed, free-text changed.
- Allow configured eSign and digital-signature integrations. For registration documents, support export/package steps required by the registration workflow without pretending that HomeFlow itself replaces statutory registration systems.

32.7 Required screens and UX

- Documents > Generate: select Project/Unit/Booking/Customer, choose applicable document family, show only valid template versions, display readiness and generate.
- Document Workspace: preview, source-data panel, clause/schedule panel, validation results, version history, approval timeline, customer-share status and execution status.
- Project > Document Factory: queue by Project showing Drafts, Awaiting Data, Awaiting Legal, Awaiting Customer, Approved for Execution, Executed/Registered and Exceptions.
- Unit/Booking/Customer 360: show current legal documents, latest approved version, pending actions and final executed copies without navigating to a separate repository.
- Admin > Template & Clause Studio: Legal/Admin can manage templates, merge-field maps, clause library, applicability rules, effective dates, approvals and retirements with sandbox/test generation before production activation.
- Bulk generation may be supported for low-risk standardized notices/letters, but legal agreements/deeds require record-level validation and configured approvals.

32.8 Permissions and segregation of duties

- Legal owns legal templates, clauses and legal deviations. Commercial/Finance owns commercial values/approved deviations. CRM initiates/generates within authority. Sales may view appropriate document readiness but must not edit legal/commercial source truth.
- Use role + Project + document-family permissions. A Project-specific Legal/CRM user sees assigned Project work; central Legal can access authorized portfolio scope.
- Sensitive identity, pricing, banking and legal fields must follow field-level permissions and masking rules in previews, exports and customer sharing.
- No user should be able to both create an unapproved deviation and self-approve it where segregation-of-duties rules require independent review.

32.9 Error prevention, audit and security

- Every generated output records template version, clause versions, data snapshot, validations, overrides, approvals, generation engine version and cryptographic checksum/hash of final artifact.
- No silent manual editing of generated DOCX/PDF outside the governed workflow should be treated as an approved executable copy. If an external edit is imported, mark it External Revision and require full comparison/reapproval.
- Maintain a complete event log for generation, validation failure, manual override, clause deviation, customer share, approval/rejection, execution, registration reference, supersession, cancellation and archive.
- Final executed documents are read-only records. Corrections after execution require a new formal corrective/addendum workflow rather than overwriting history.

32.10 AI-enabled sophistication

- Document Copilot can answer: What data is missing? Why is this deed blocked? What changed from v2 to v3? Which clauses differ from the approved project standard? Which agreements are waiting on customer/legal action?
- AI can extract structured data from received legal documents and compare it with HomeFlow master data, flagging mismatches for human review.
- AI can summarize lengthy negotiated redlines and classify risk/severity, but it must cite the exact changed clause/value and cannot autonomously approve legal deviations.

- AI can propose the next best action and prepare customer/legal communication drafts subject to role-appropriate approval.

32.11 Minimum acceptance tests for Emergent

- For an eligible booked Unit, CRM selects AOS and the system offers only the currently approved template(s) applicable to that Project/property/transaction context.
- Generation is blocked when a mandatory applicant/commercial/unit field is missing; the user can click the error and navigate to the authoritative source record.
- A valid AOS is generated with applicant names, Unit details, consideration, schedules and clauses populated without manual re-entry and with zero unresolved merge tokens.
- Changing the approved consideration after Draft v1 does not mutate v1; generating v2 uses a new data snapshot and comparison clearly shows the change.
- A CRM user cannot edit a Locked legal clause. A Legal-approved deviation creates a new version and retains the original clause/version in audit history.
- A Sale Deed cannot move to Approved-for-Execution until configured legal/commercial/registration prerequisites and approvals are complete.
- A final executed/registered document is checksum-identified, locked and visible from Project, Unit, Booking and Customer records to authorized users.
- A retired template cannot be used for new generation after its effective end date, but historic documents generated from it remain fully viewable and auditable.
- Project-specific and central Legal users see the correct queues according to Project Team Assignment and document permissions.
- Lease Agreement generation supports the same governed model while using lease-specific parties, commercial terms, schedules and approval rules rather than sale-only assumptions.

Appendix A. Recommended Core Status Taxonomy

Object	Recommended statuses
Action	New; In Progress; Waiting Internal; Waiting Customer; Blocked; Ready for Approval; Closed; Cancelled
Commitment	Draft; Approved; Active; At Risk; Fulfilled; Breached; Waived/Cancelled
Document	Required; Requested; Received; Validating; Accepted; Rejected; Superseded; Expired
Snag	Open; Assigned; In Progress; Ready for QA; Reopened; Verified; Closed
Registration	Not Ready; Readiness In Progress; Ready; Slot Booked; Completed; Document Pending
Handover	Not Eligible; At Risk; Eligible; Appointment Booked; In Progress; Completed; Reopened
Customer Health	Healthy; Watch; At Risk; Critical

Object	Recommended statuses
Financial Risk	Low; Watch; At Risk; Disputed; Default/Legal
Unit Change / Customisation	Draft; Requested; Feasibility Review; Costing; Awaiting Approval; Awaiting Customer; Awaiting Payment; Approved; Released; In Progress; Ready for QA; QA Verified; Customer Accepted; As-Built Closed; Rejected; Withdrawn; Cancelled
Changeability Gate	Open; Closing; Conditional; Exception Only; Hard Closed
Change Window Hold	Requested; Project Review; Approved; Active; Expired; Released; Rejected; Cancelled

Appendix B. Minimum Event Log

- Booking created / revised / cancelled / transferred.
- Sales handover submitted / returned / accepted.
- Demand raised / receipt posted / reversal / waiver / TDS verified.
- Collection forecast created / revised / snapshot locked / probability changed / expected date changed / scenario changed.
- Project team assigned / reassigned / shared scope changed / effective-dated ownership changed.
- Document requested / received / accepted / rejected / superseded.
- Agreement generated / revised / approved / executed.
- Registration readiness achieved / slot booked / completed.
- Unit readiness component passed / failed / reverified.
- Snag created / assigned / rectified / verified / reopened / closed.
- Commitment created / approved / at risk / fulfilled / breached.
- Customer contact sent / response received / sentiment changed.
- Escalation created / upgraded / recovery plan / closed.
- Handover eligibility reached / blocked / appointment / completed.
- Unit change requested / feasibility assessed / quote issued / customer approved / payment cleared / drawing released / execution started / QA verified / customer accepted / as-built closed.
- Unit physical progress updated / bulk-applied / unit exception recorded / corrected / published to Sales-CRM / freshness threshold breached / verification requested.
- Changeability gate opened / closing forecast changed / restricted / exception-only / hard closed / reopened by authorized physical correction.
- Change Window Hold requested / approved / activated / expired / released / rejected.
- Prospect personalisation requirements captured / unit-match generated / unit compared / selected.
- Warranty/service case opened / resolved / reopened.
- Document template created / approved / activated / retired.
- Generated document created / validation failed / revised / superseded.
- Clause selected / deviation requested / deviation approved or rejected.
- Document shared with customer / customer comments received / customer accepted.
- Document approved for execution / eSigned / wet-signed / registered / archived.
- External revision imported / compared / reapproved or rejected.

33. Emergent Build Instructions - Project-Owned Live Unit Status for Sales & CRM

R1/R2 implementation rule: build the Project Unit Status Console and Unit Digital Twin so Project/Construction users can operate them directly in HomeFlow. Do not make any external construction platform a prerequisite. Generic integration primitives may be added later only when justified by an approved connector requirement.

This is a mandatory closed-loop operating capability. Project/Construction owns the physical truth of each Unit. Sales and CRM consume that truth in real time while interacting with prospects and customers. Sales/CRM must never directly edit construction progress, QA truth, technical feasibility or hard changeability gates.

33.1 Project Unit Status Console

Provide a Project -> Phase/Tower/Block -> Floor/Zone -> Unit interface where authorized Project users can update component-level physical status, percent complete where meaningful, actual/start/forecast dates, QA status, evidence/photos, expected readiness, next planned activity and remarks. Support bulk updates across selected units with an affected-unit preview, while allowing controlled unit-level exceptions.

The console must show data freshness and source for every status: HomeFlow Project-team update, approved bulk update, QA evidence, procurement/material update recorded in HomeFlow, authorized correction, or an optional generic external-system event if configured. HomeFlow Project updates must publish to the Unit Digital Twin and trigger recalculation of mapped changeability gates without requiring any external system.

33.2 Sales / CRM Read-Only Unit Intelligence

On Inventory, Unit 360, opportunity and customer screens, expose the latest Project-owned unit status: overall construction stage, component/trade status, latest evidence date, expected readiness/handover window, QA/readiness status, Customisation Flexibility score, gate summary and gates closing soon. Show Last Project Update and a freshness indicator prominently.

If the source data exceeds the configured freshness threshold, display Verification Required and prevent the interface from presenting technical changeability as a confident promise. Sales/CRM may request clarification from Project but cannot overwrite the physical state.

33.3 Sales Interaction and Unit Selection

Sales must be able to filter and compare available units using both commercial attributes and live changeability. Filters include project, unit type, price, possession window, construction stage, Highly Customisable, Layout Flexible, Kitchen Changes Open, Electrical Changes Open, Flooring Selection Open, Bathroom Specification Open and other configured categories.

Capture prospect requirements as Must Have, Preferred and Not Important. Rank available units against those requirements using current Project-owned gate states. Explain every match or mismatch. A compatibility result is sales guidance only and must not be represented as final engineering approval.

33.4 Event Propagation

A Project progress event must propagate through the platform: HomeFlow Project update -> Unit Digital Twin -> component/trade state -> changeability rule evaluation -> Sales/CRM inventory visibility -> affected prospect/customer warnings -> Change Request routing where applicable. If an optional external system is

connected later, it may submit the same normalized event through a generic adapter, but HomeFlow must remain fully functional without it and must not require duplicate manual entry.

Example: when electrical first-fix starts for Floors 8-12, affected units automatically move mapped electrical gates from Open to Conditional/Exception according to policy; Sales sees the new status immediately and active prospects with electrical Must Have requirements are flagged for review.

33.5 Ownership, Permissions and Accountability

Project/Construction owns physical progress and forecast readiness. Design/Engineering owns technical feasibility rules. QA owns inspection truth. Procurement contributes order/material events. Sales has read/use rights for prospect conversations. CRM has read/use rights and may initiate Change Requests. Commercial owns pricing/margin treatment. Management controls designated exceptions. Hard safety/statutory gates remain non-overridable.

33.6 Minimum Acceptance Tests for Emergent

1. A Project user updates a unit component once and the new status is visible on Sales and CRM views without duplicate entry.
2. A bulk Project update can affect many units while preserving authorized unit exceptions.
3. Sales cannot edit Project-owned physical status or technical gates.
4. Sales can compare available units by live customisation/changeability and prospect requirements.
5. Stale Project data displays Verification Required before Sales/CRM relies on it.
6. A mapped construction event automatically recalculates relevant gates and records the source event.
7. Active prospects/customers affected by a gate transition can be identified and routed for follow-up.
8. Every physical-status correction records actor, timestamp, prior value, new value and reason.

34. Emergent Build Instructions - Configurable Project Journey & Universal Timeline/SLA Engine

This section is a core platform requirement. Do not hardcode any Project-specific number of days, dates, charges, milestones, stage names or customer wording into application logic. HomeFlow must provide a generic lifecycle engine configured through data and inherited Project templates. Existing Project SOPs are inputs to configuration and continuous-improvement review, not software source code.

34.1 Configuration hierarchy

- Pranava Standard Journey Template: enterprise default lifecycle and common terminology.
- Project Template: inherits the standard and applies authorized Project-specific stages, tasks, durations, gates, documents, owners, communications and visibility rules.
- Unit / Booking instantiation: when a booking is confirmed, create the applicable journey version while preserving links to the Unit Digital Twin and Project template version.
- Exceptional override: authorized record-level change to plan, owner, gate or sequence with reason, approver and audit trail; never silently mutate the template.

34.2 Generic lifecycle and parallel-stream model

Lifecycle stage	Core intent	Important parallel streams
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0 Unit / Pre-Sales	Expose reliable unit truth before sale.	Construction progress, expected readiness, changeability, inventory.
1 Booking	Create commercial relationship.	Booking payment, applicant capture, approvals, CRM handover.
2 Funding	Establish funding path.	Loan/self-funding, bank readiness, future disbursement.
3 Agreement & Documentation	Complete contractual package.	KYC, AOS/agreement generation, approvals, TDS/documents.
4 Construction & Unit Journey	Maintain physical truth.	Progress, evidence, customer updates, customisation/change requests.
5 Demands & Collections	Convert milestones into cash.	Demand, receipt, TDS, loan disbursement, collection forecast.
6 Pre-Registration	Converge registration prerequisites.	Finance, legal, documents, statutory, customer availability.
7 Registration	Complete legal transfer.	Sale deed, challan, slot, registration evidence.
8 Pre-Handover	Converge readiness.	QA, snags, finance/legal closure, commitments, inspection.
9 Handover	Execute possession.	Keys, documents, warranties, readings, FM orientation.
10 Post-Handover	Support occupancy and long-term relationship.	DLP/warranty, service, FM, home passport, advocacy.

The engine must allow parallel streams. A journey is not blocked merely because a prior numbered stage is open unless an explicit dependency or gate says so. Numbering is primarily for customer/management comprehension; dependency rules govern actual execution.

34.3 Universal date model

Field	Requirement
Planned duration	Configurable working/calendar-day duration inherited from policy; optional where milestone/event-based.
Baseline planned start/end	Original approved plan captured when the journey/task is instantiated. Immutable except controlled baseline reset.
Current planned start/end	Latest approved operational plan after authorized revisions.
Forecast start/end	Best current expectation based on progress, blockers and owner input/system calculation.

Actual start/end	System/manual evidence of real commencement and completion.
Variance	Forecast vs current plan, actual vs baseline/current plan, and SLA variance.
Calendar	Project/company calendar, weekends, holidays and configured working hours.
Pause / hold	Permitted only for configured reasons with start/end, owner and evidence.
Delay reason	Structured root-cause code plus optional narrative; mandatory beyond configured tolerance.
Confidence	Optional confidence for forecast dates/windows; explain source/driver when system-generated.

34.4 SLA versus plan

SLA and plan are related but separate. SLA answers how long a service/process is allowed to take; Plan answers when this specific record is intended to occur. A task may be within SLA but late to the Project plan, or vice versa. HomeFlow must calculate both independently.

- SLA policy fields: duration, calendar, start trigger, pause reasons, warning threshold, breach threshold, escalation route.
- Plan policy fields: baseline date/duration, start trigger, dependency logic, current plan revision rules, forecast rules and variance tolerance.
- Status derivation: On Track, Due Soon, At Risk, Overdue, Completed On Time, Completed Late, Paused/Waiting with reason.
- Forecast changes must not overwrite baseline or prior forecasts; maintain a dated revision history for predictability analysis.

34.5 Required screens

- Admin > Journey Template Studio: create/version Pranava Standard and Project templates, stages, sub-stages, dependencies, gates, durations, owners, customer visibility and notifications.
- Project Journey Control: all active Project journeys with current stage/milestone, next planned event, forecast variance, SLA health and blockers.
- Customer/Booking Journey Timeline: customer-friendly milestone layer plus internal detailed layer with plan/current/forecast/actual dates.
- Stage/Task Detail: owner, plan, forecast, actual, SLA clock, dependencies, blockers, documents, comments, evidence and change history.
- Management Analytics: Project/stage bottlenecks, plan vs actual, forecast accuracy, SLA performance and root-cause trends.

34.6 Required data objects / fields

- JourneyTemplate, JourneyTemplateVersion, JourneyStageTemplate, JourneyTaskTemplate and dependency/gate definitions.
- JourneyInstance and StageInstance keyed to Project, Unit, Booking and Customer(s) as applicable.
- TimelinePlanRevision: baseline/current plan revisions with effective date, actor, reason and approval.

- TimelineForecastRevision: forecast history with source, confidence and driver/blocker references.
- SlaPolicy and SlaClockEvent: start, pause, resume, warning, breach, completion.
- ProjectCalendar / WorkingCalendar with holidays and working-day rules.
- DelayReason taxonomy and Project/process-specific allowed codes.

34.7 Minimum acceptance tests for Emergent

1. A Project can inherit the Pranava Standard Journey and override a configured duration without code change.
2. Two Projects can use different planned durations/SLAs for the same generic stage while using the same application code.
3. A journey displays baseline, current plan, forecast and actual dates without overwriting prior revisions.
4. SLA health can differ from Project plan health and both are explained correctly.
5. Changing a Project template version does not silently alter already active journeys unless an approved migration rule is executed.
6. Construction, collections and customisation work can run in parallel; only explicit dependencies/gates block progression.
7. A delayed forecast marks the journey At Risk before the planned end date and identifies the responsible blocker/owner.
8. Plan and SLA analytics roll up by Project, stage, department, role and owner.
9. Customer-facing journey hides internal-only tasks but shows approved milestone/status/timeline information.
10. All overrides, date revisions, pauses, template changes and completion events are auditable.

35. Illustrative Project Mapping - East Crest as Configuration, Not Code

The East Crest CRM roadmap is used as a real-world validation case. Its current seven-stage operating sequence maps cleanly into the generic HomeFlow lifecycle, but its stated day counts, notices, charges, approvals and wording must remain configurable Project policy. This example validates the generic design and helps optimize the SOP; it must not create East Crest-specific application branches.

Current East Crest concept	Generic HomeFlow mapping	Optimization in HomeFlow
Flat Selection & Booking Allotment	1 Booking & Allotment	Add Unit pre-sales truth before booking; structured sales-to-CRM handover, plan/actual dates and evidence.
Financial Sanction & Loan Approval	2 Funding & Financial Setup	Loan/self-funding workflow can run in parallel; track forecast disbursement and blockers.
Agreement of Sale	3 Agreement & Documentation	Use Document Factory, approvals, versioning, TDS/document checklist and configurable SLA/plan.
Construction Stage Demands & Bank Funds	4 Construction + 5 Demands & Collections	Separate physical progress from financial demand while linking

		milestone events; improve project cash forecast.
Pre-Registration Financial Clearance	6 Pre-Registration Readiness	Gate combines finance, documents, legal/statutory and customer readiness.
Sale Deed & Sub-Registrar Slot	7 Registration	Document generation, validation, challan/slot, execution and evidence with configurable plan/SLA.
Pre-Handover Inspection & Key Takeover	8 Pre-Handover + 9 Handover	Add Unit Readiness, QA/snags, commitments, handover gate, scheduling, possession evidence and FM transition.
Not explicit in current high-level SOP	0 Unit / Pre-Sales + 10 Post-Handover	Add pre-sale unit/changeability intelligence and post-handover DLP/warranty/facilities lifecycle.

Governance rule: Project teams and functional owners may propose Project-specific template changes, but the standard Pranava lifecycle vocabulary should remain stable wherever possible so portfolio analytics and cross-project benchmarking remain meaningful.

35.1 Additional mandatory audit events

- Journey template/version created, approved, activated, superseded.
- Project template inherited/overridden; override approved/rejected.
- Baseline plan created/reset; current plan revised.
- Forecast revised; confidence/source changed.
- SLA clock started/paused/resumed/warned/breached/completed.
- Dependency/gate opened/closed/waived/overridden.
- Delay reason recorded/changed; customer-visible milestone/date changed.